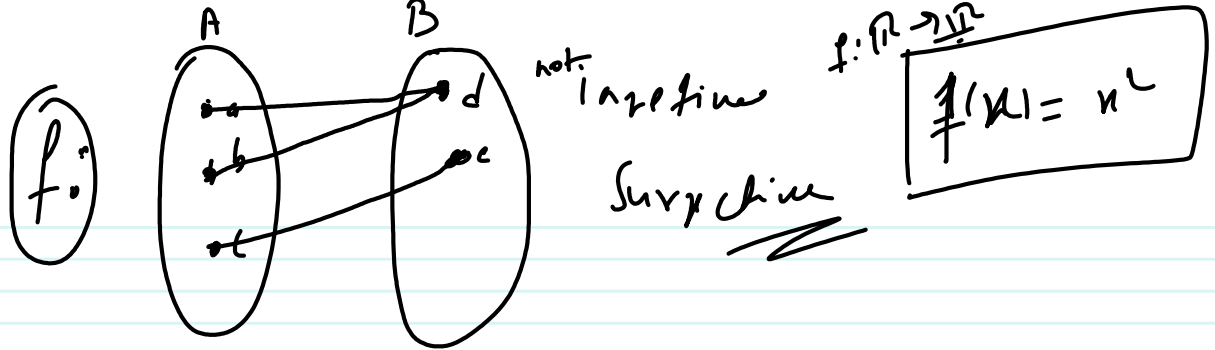




$$f \subseteq A \times B \quad f = \{ (a, d), (b, d) \} \subseteq f$$

$$S = \{ a, b, c, d \}$$

$$R = \{ (a, b), (b, c), (a, c), (d, d) \}$$

$$R = \{ (a, b), (b, c), (a, c) \}$$

$(a, b), (b, c) \Rightarrow (a, c)$

$(a, b), (b, a) \Rightarrow (a, a)$

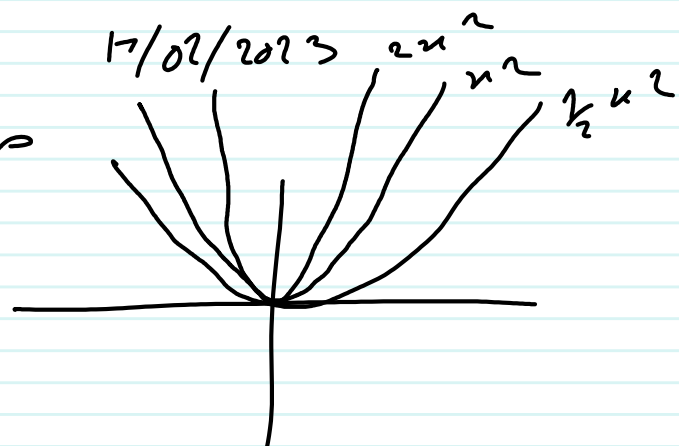
$(a, b), (b, c), (c, a) \Rightarrow (a, a)$

Math. 1

$$P_1(x) = 1x^2$$

$$P_1(x) = \frac{1}{2}x^2$$

$$P_2(x) = 2x^2$$



S, R , $(a, b) \in R \Rightarrow (b, a) \in R$ symmetric

$$(a, b) \in R, (b, a) \in R \Rightarrow a = b$$

$$(a,b) \in R, (b,a) \in R \Rightarrow$$

$$A = \{a,b,c\}, B = \{a,b,c,d\}$$

$$A \subseteq B, B \not\subseteq A$$

$$A \times B$$

$$(A \times B) =$$

$$R = \{ (A,B) \mid A \subseteq B \}$$

$$(a,b) \in R \wedge (b,a) \in R \Rightarrow a=b \quad \text{Antisym}$$

$$(I) \quad (a,b) \in R \quad \text{Antisymmetrie}$$

$$(II) \quad (a,b) \in R, (b,a) \in R \Rightarrow (a=b) \quad \text{Antisym}$$

$$A \subseteq B \quad (A,B) \in R \quad \text{Antisym}$$

$$(A,B) \in R \wedge (B,A) \in R$$

$$A \subseteq B$$

$$B \subseteq A$$

$$B \subseteq A$$

$$\Rightarrow A=B$$

$$R = \left\{ \left(\underset{A}{\{1,2\}}, \underset{B}{\{1,2,3\}} \right), \left(\underset{C}{\{1\}}, \underset{D}{\{1,2\}} \right), \left(\underset{A}{\{1,2,3\}}, \underset{B}{\{1,2\}} \right) \right\}$$

$$A \subseteq B$$

$$C \subseteq B$$

$$A \subseteq B$$

$$B \subseteq A$$

$$A=B$$

$$B \subseteq A$$

$$S = \{ \}$$

$$S = \{ \text{family} \}$$

$$R = \{ (A,B) \mid B \text{ is el. bro } A \}$$

$$= (A,B)$$

$$(B,A)$$

$$B \cap A \in R \quad A=B$$

$$P(x) = x^3$$



$$x=0$$

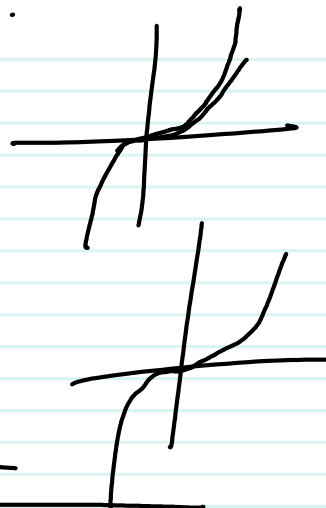
$$x^3$$

at

$$P(0) = 0$$

$$P(0.000001) = (0.000001)^3 = \text{positi}$$

$$P(-0.000001) = (0.000001)^3 = \text{neg}$$



$$\underline{\underline{P(x)}}$$

$$= \frac{\text{even de. poly}}{\text{even fun}}$$

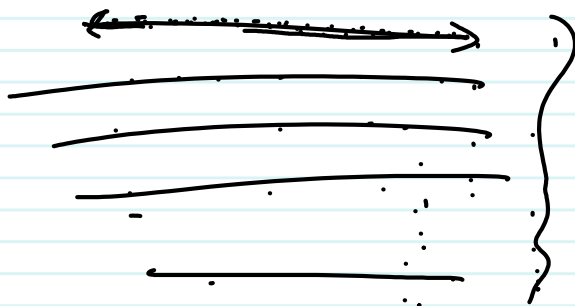
$$P(x) = \overbrace{ax^3}^{\text{odd deg}} + bx^2$$

$$ax^4 + \dots$$

✓
✓
✓
(-1)

even fun

$$\text{even } f(-x) = f(x) \quad \forall x \in \text{domain}$$



$$(3x(\text{norm}) - 1)$$

$$\underline{x \rightarrow \infty}$$

$$P(x) = \dots \infty$$

$$P(x) = a \textcircled{3} + bx^6 + c$$

$$P_1(n) = \overset{a>0}{\underline{5}} n^{\overset{odd}{3}} + \dots \quad P_3 = \overset{n \rightarrow \infty}{\underline{2}} n^{\overset{4}{4}} + \dots \quad P_3(n) \rightarrow +\infty$$

$$P_2(n) = \overset{a<0}{\underline{-5}} n^{\overset{odd}{3}} + \dots \quad P_4(n) = \overset{n \rightarrow -\infty}{\underline{-2}} n^{\overset{4}{4}} \quad P_4(n) \rightarrow -\infty$$

$$P(n) = \overset{\uparrow}{\underline{-n^3}} + \underline{2n}$$

$$P(2) = -4 + \underline{2 \times 2}$$

$$P(3) = -9 + \underline{2 \times 3} = \underline{-3}$$

$$n \rightarrow +\infty$$

$$n \rightarrow -\infty$$

$$n \rightarrow +\infty$$

$$n \rightarrow -\infty$$

$$P_2(n) \rightarrow -\infty$$

$$P_2(n) \rightarrow +\infty$$

$$P_1(n) \rightarrow \infty$$

$$P_1(n) \rightarrow -\infty$$

$$\sqrt[n]{n^{16}} \quad \text{X}$$

$$\sqrt{(2^4)} = \sqrt{16} = 4$$

$$\sqrt{(2^4)} = (2^4)^{\frac{1}{2}} = 2^{4 \times \frac{1}{2}} = 2^2 = 4$$

$$\boxed{A \cup B}$$

$$\frac{7+3\sqrt{5}}{2} = \frac{7}{2} + \frac{3\sqrt{5}}{2}$$

$$\frac{A \cup B}{\text{slope}}$$

$$A \cap B$$

$$A \text{ and } B$$

$$\underline{\text{slope}}$$

$$\text{run}$$

$$\text{mat}$$

$$\text{start} \rightarrow$$

$$()$$

$$()$$

$$()$$

